ME675: Predictive Control of Dynamic Systems

Introduction to Model Predictive Control

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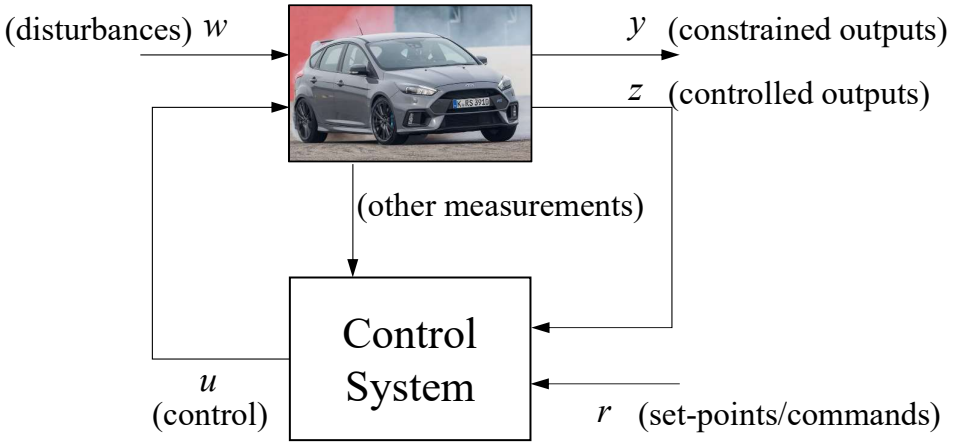
Systems with Constraints

Most systems have constraints that must be enforced during system operation:

- Physical constraints, e.g. actuator limits
- Performance constraints, e.g. overshoot
- Safety constraints, e.g. temperature/pressure limits

Operating points are often near constraints.

- It is a must to satisfy hard constraints strictly.
- It is desirable to satisfy soft constraints whenever possible.



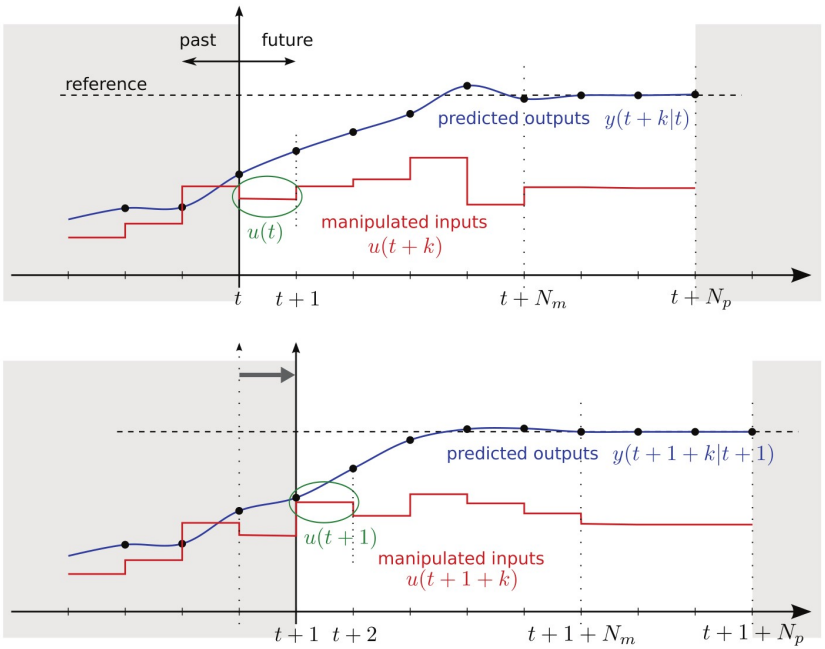
Constrained Control Methods

Constrained control is concerned with the development of control algorithms with the capability of enforcing constraints.

Several classes of constrained control techniques have been developed:

- Re-tuning the controller to avoid constraint violation
- Anti-windup for systems with control constraints/saturation
- Control input saturation tolerant design
- Barrier functions
- Invariance-based techniques
- **Model Predictive Control**
- ...

Model Predictive Control (Receding Horizon Control)



- MPC exploits a system **model** to **predict** the future response and to accordingly choose the **best control** action.
- The **best control** action is achieved by performing **optimization** against a performance objective and constraints involving system states, inputs and outputs.

Why Predictive Control?

The use of prediction seems logical: humans do this naturally. Some main advantages are:

1. Intuitive concept, easy to understand and implement for a variety of systems.
2. Systematic handling of constraints.
3. Handles Multi-input-multi-output systems.
4. Make good use of future target information implicitly.
5. Handles challenging dynamics (unlike PID).

MPC History

- Propoi (1963) proposed a form of MPC utilizing linear programming, for the control of linear systems with hard constraints on the control.
- Lee and Markus (1967) pointed out the possibility of MPC in their book on optimal control.
  One technique for obtaining a feedback controller synthesis is to measure the current control process state and then compute very rapidly the open-loop control function. The first portion of this function is then used during a short time interval after which a new measurement of the process state is made and a new open-loop control function is computed for this new measurement. The procedure is then repeated.
- A big surge in interest in MPC occurred when Richalet, Rault, Testud, and Papon (1978) advocated its use for process control.
- MPC soon became the most useful method in modern control technology for control problems with hard constraints - with thousands of applications.
- In the last two-three decades theoretical foundation for MPC has been established providing this originally heuristic technique with solid theoretical foundation.
- A lot of algorithms for "fast MPC" have been developed so that the technology can be applied to systems requiring very fast sampling rates.

MPC in Industry

TABLE 1 A list of the survey results in order of industry impact as perceived by the committee members.

Rank and Technology	High-Impact Ratings	Low- or No-Impact Ratings
PID control	100%	0%
Model predictive control	78%	9%
System identification	61%	9%
Process data analytics	61%	17%
Soft sensing	52%	22%
Fault detection and identification	50%	18%
Decentralized and/or coordinated control	48%	30%
Intelligent control	35%	30%
Discrete-event systems	23%	32%
Nonlinear control	22%	35%
Adaptive control	17%	43%
Robust control	13%	43%
Hybrid dynamical systems	13%	43%

T. Samad, "A Survey on Industry Impact and Challenges Thereof", IEEE Control Systems Magazine, vol. 31, no. 1, 2017, pp. 17–18

MPC in Industry

Table 2
The percentage of survey respondents indicating whether a control technology had demonstrated ("Current Impact") or was likely to demonstrate over the next five years ("Future Impact") high impact in practice.

	Current Impact	Future Impact
Control Technology	%High	%High
PID control	91%	78%
System Identification	65%	72%
Estimation and filtering	64%	63%
Model-predictive control	62%	85%
Process data analytics	51%	70%
Fault detection and identification	48%	78%
Decentralized and/or coordinated control	29%	54%
Robust control	26%	42%
Intelligent control	24%	59%
Discrete-event systems	24%	39%
Nonlinear control	21%	42%
Adaptive control	18%	44%
Repetitive control	12%	17%
Hybrid dynamical systems	11%	33%
Other advanced control technology	11%	25%
Game theory	5%	17%

Table 3
Assessments of the current impact of four advanced control technologies by respondents with different industry backgrounds.

	MPC		Robust Control		Adaptive Control		Nonlinear Control	
	%High	%Low/None	%High	%Low/None	%High	%Low/None	%High	%Low/None
Process Industry	59	9	9	44	9	50	9	56
Aerospace	64	14	36	14	29	36	36	21
Automotive	60	10	10	30	30	50	20	30

T. Samad, M. Bauer and S. Bortoff et al. "Industry engagement with control research: Perspective and messages", Annual Reviews in Control, 49 (2020), 1-14

Mass Production of MPC in Automotive Application

ODYS AND GM BRING ONLINE MPC TO PRODUCTION!

POSTED ON **April 10, 2018** AUTHOR **Daniele Bernardini** CATEGORIES **Automotive, Projects**

ODYS is proud to announce that the **model predictive control** system developed with **General Motors** for **engine and powertrain control is in production** since 2018.

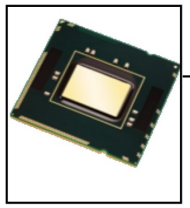
The control design concept has been presented on April 10th, 2018 at SAE WCX World Congress Experience in Detroit, MI.

Read more and check the SAE papers here:

- **MPC for turbocharged gasoline engines**
- **Supervisory MPC for powertrain with continuously variable transmission (CVT)**

Source: https://www.odys.it/odys-and-gm-bring-online-mpc-to-production/

MPC Applications



Computer control — ns

μs — Power systems

Traction control — ms

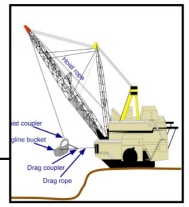
Seconds — Buildings

Refineries — Minutes

Hours — Nurse rostering

Train scheduling — Days

Weeks — Production planning



Source: Prof. Manfred Morari

Why so popular in industry?

It has been proven to improve profits by giving superior control compared to classical techniques.

1. If one is confident that the variance of the output can be reduced, one can then safely operate closer to a constraint and therefore increase output quantity or quality.
2. The ability to incorporate constraints explicitly enables **optimum** constrained performance as opposed to the consequences of **ad hoc** fixes.

Basic Ingredients of Model Predictive Control

MPC is an optimization-based control. The basic concept of MPC is to solve a (deterministic) optimization problem against a performance objective and constraints involving system states, inputs and outputs.

For formulation of an Optimal Control Problem (OCP), the followings are needed:

1. Equations of Motions: model
2. Cost functional (or performance index): incremental cost, terminal cost
3. Initial conditions
4. Terminal conditions
5. Constraints: state, control, mixed state and control

OCP Formulation

- Equations of Motion

$$\dot{x} = f(x, u)$$

$x \in \mathbb{R}^n$: the state vector; $u \in \mathbb{R}^m$: the control vector.

- Cost functional being minimized (or maximized)

$$J = K\left(x(t_f), t_f\right) + \int_{t_0}^{t_f} L\left(x(t), u(t)\right) dt$$

K: the terminal cost; L: the incremental cost; t_f fixed or free.

- Constraints

Initial and terminal conditions: $x(t_0) = x_0, x(t_f) = x_f$
State constraints: $x_{\min} \leq x(t) \leq x_{\max}$
Control constraints: $u_{\min} \leq u(t) \leq u_{\max}$
Mixed constraints: $g(x(t), u(t), t) \leq 0$

OCP Formulation

By solving an Optimal Control Problem, you can determine:

- control $u(t)$ as a function of time,
- terminal time t_f (if the terminal time is free),
- and the associated state trajectory $x(t)$,

which minimizes the cost functional.

In this course, the **discrete-time optimal control problem** will be discussed because the discrete-time OCP can be viewed as a **finite-dimensional optimization problem**. The optimization variable consists of the control inputs as a finite number of time instances.

Finite-Dimensional Optimization Problem

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & f(x) \\ \text{subject to} \quad & x \in \mathcal{D} \subseteq \mathbb{R}^n \end{aligned}$$

- $f : \mathcal{D} \to \mathbb{R}$ is the objective (or cost) function and real-valued
- $x \in \mathbb{R}^n$ are the decision variables
- $\mathcal{D} \subseteq \mathbb{R}^n$ is the constraint set
  - ✓ $\mathcal{D}$ is the domain over which f is defined
  - ✓ $\mathcal{D}$ is determined by inequality constraints of the problem
- The optimization problem is feasible if $\mathcal{D}$ is not empty, and $x \in \mathcal{D}$ is a feasible point
- If $\mathcal{D} = \mathbb{R}^n$, the problem is an unconstrained optimization problem; otherwise, it is a constrained optimization problem

Finite-Dimensional Optimization Problem

$$\begin{aligned} \underset{x}{\text{minimize}} \quad & f(x) \\ \text{subject to} \quad & x \in \mathcal{D} \subseteq \mathbb{R}^n \end{aligned}$$

- A feasible point $x^* \in \mathcal{D}$ is an optimal point if

$$f(x^*) \leq f(x), \forall x \in \mathcal{D}$$

- $f^* = f(x^*)$ is the optimal value

✓ Goal: Find an optimal point that minimizes the objective function over the constraint set $\mathcal{D}$

Mathematical Formulation

$$
\begin{aligned}
\min_{x} \quad & f(x) \\
\text{s.t.} \quad & x \in \mathcal{D} \subseteq \mathbb{R}^n \\
& h_i(x) \leq 0, \ \forall i = 1, \ldots, m \\
& g_j(x) = 0, \ \forall j = 1, \ldots, l
\end{aligned}
$$

In most cases, the constraint set $\mathcal{D}$ can be broken into explicit constraints that are written in terms of other functions with equalities and inequalities, and other constraints that are not so explicit.

- $h_i : \mathbb{R}^n \to \mathbb{R}, \forall i = 1, \ldots, m$
- $g_j : \mathbb{R}^n \to \mathbb{R}, \forall j = 1, \ldots, l$
- The feasible set: $Y = \{x \in D | \, h_i(x) \leq 0, g_j(x) = 0, \ \forall i, j\}$

✓ Maximizing $f(x)$ is equivalent to minimizing $-f(x)$

Major Issues

Major issues of the optimization problem include

- Existence: When does there exist an optimal solution to the problem?
- Uniqueness: If the optimal solution exists, is it unique?
- Characterization: Are there analytical tools to characterize the optimal solution?
- Algorithms: If the analytical approach is not applicable, what numerical techniques should we use to solve the problem?

Infimum

If a set of real numbers $\mathcal{S} \subset \mathbb{R}$ is bounded from below,
i.e., there exists $\underline{\alpha} \in \mathbb{R}$ such that $\underline{\alpha} \leq x$ for all $x \in \mathcal{S}$,
this set has a **greatest lower bound** or an **infimum**, α_*, which satisfies:

- $\forall x \in \mathcal{S}, x \geq \alpha_*$,
- $\forall \epsilon > 0, \exists x \in \mathcal{S}$ such that $x - \epsilon < \alpha_*$

The infimum α_* of $\mathcal{S}$ is written as $\alpha_* = \inf \mathcal{S}$. If $\alpha_* \in \mathcal{S}$, then α_* is the minimum of $\mathcal{S}$, $\alpha_* = \min \mathcal{S}$.

Examples: $\mathcal{S} = [0, 1)$, $\mathcal{S} = (0, 1)$, $\mathcal{S} = \mathbb{R}$

Supremum

If a set of real numbers $\mathcal{S} \subset \mathbb{R}$ is bounded from above,
i.e., there exists $\overline{\alpha} \in \mathbb{R}$ such that $\overline{\alpha} \geq x$ for all $x \in \mathcal{S}$,
this set has a **least upper bound** or an **supremum**, α^*, which satisfies:

- $\forall x \in \mathcal{S}, x \leq \alpha^*$,
- $\forall \epsilon > 0, \exists x \in \mathcal{S}$ such that $x + \epsilon > \alpha^*$

The supremum α^* of $\mathcal{S}$ is written as $\alpha^* = \sup \mathcal{S}$. If $\alpha^* \in \mathcal{S}$, then α^* is the maximum of $\mathcal{S}$, $\alpha^* = \max \mathcal{S}$.

Examples: $\mathcal{S} = [0, 1]$, $\mathcal{S} = (0, 1)$, $\mathcal{S} = \mathbb{R}$

Vector Space

Vector space = linear space = linear vector Space

A linear space over a field $\mathbb{F}$, $(\mathbb{V}, \mathbb{F})$, consists of a set $\mathbb{V}$ of vectors, a field $\mathbb{F}$, and two operations, vector addition and scalar multiplication.

Addition and scalar multiplication obey the following rules:

- for $x, y \in \mathbb{V}$, $x + y = y + x$ (commutativity);
- for $x, y, z \in \mathbb{V}$, $x + (y + z) = (x + y) + z$ (associativity);
- there exists a null vector, $0 \in \mathbb{V}$ such that $0 + x = x + 0 = x$;
- for each vector $x \in \mathbb{V}$, there exists an inverse $-x \in \mathbb{V}$ such that $x + (-x) = 0$;
- for $\alpha \in \mathbb{F}$ and $x, y \in \mathbb{V}$, $\alpha(x + y) = \alpha x + \alpha y$;
- for $\alpha, \beta \in \mathbb{F}$ and $x \in \mathbb{V}$, $(\alpha + \beta)x = \alpha x + \beta x$;
- for $\alpha, \beta \in \mathbb{F}$ and $x \in \mathbb{V}$, $\alpha(\beta x) = (\alpha\beta)x$;

Vector Space

Vector space = linear space = linear vector Space

A linear space over a field $\mathbb{F}$, $(\mathbb{V}, \mathbb{F})$, consists of a set $\mathbb{V}$ of vectors, a field $\mathbb{F}$, and two operations, vector addition and scalar multiplication.

Examples

- $(\mathbb{F}^n, \mathbb{F})$ where $\mathbb{F}^n = \mathbb{F} \times \mathbb{F} \times \cdots \times \mathbb{F}$
- $(\mathbb{R}^n, \mathbb{R})$, $(\mathbb{C}^n, \mathbb{C})$, $(\mathbb{C}^n, \mathbb{R})$
- $(\mathbb{R}^n, \mathbb{C})$ is not a vector space. Why? $(1 + j) * 1 = 1 + j \notin \mathbb{R}$

In this course, the n-dimensional real vector space, $(\mathbb{R}^n, \mathbb{R})$, will be considered.

Norm

A function $||x|| : \mathbb{R}^n \to \mathbb{R}$ is said to be a norm if the following properties hold

- $||x|| \geq 0$ and $||x|| = 0 \Leftrightarrow x = 0$
- $||\lambda x|| = |\lambda| \cdot ||x||$, $\lambda \in \mathbb{R}$ (absolute homogeneity)
- $||x + y|| \leq ||x|| + ||y||$ (triangular inequality)

Examples

- 2-norm: $||x||_2 = \sqrt{x^T x} = \left(\sum_{i=1}^{n} x_i^2 \right)^{1/2}$
- 1-norm: $||x||_1 = \sum_{i=1}^{n} |x_i|$
- Infinity-norm: $||x||_\infty = \max\{|x_1|, |x_2|, \cdots, |x_n|\}$
- p-norm: $||x||_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$

Norm

Fact: If $||\cdot||$ and $||\cdot||'$ are two norms on $\mathbb{R}^n$, then there exists constants $c_1, c_2 > 0$ such that

$$c_1||x||' \leq ||x|| \leq c_2||x||' \quad \forall x \in \mathbb{R}^n$$

The ratio between two norms is bounded below and above, independent of x

The above fact implies that in the finite-dimensional space, we can use any norm to define convergence and continuity on $\mathbb{R}^n$

Normed Balls

An open ball $\mathcal{B}_\epsilon(x_0)$ with a radius of $\epsilon > 0$ is a subset of $\mathbb{R}^n$, defined as

$$\mathcal{B}_\epsilon(x_0) = \{x \in \mathbb{R}^n : ||x - x_0|| < \epsilon\}.$$

Note that $\mathcal{B}_\epsilon(x_0)$ depends on the norm $|| \cdot ||$ used.

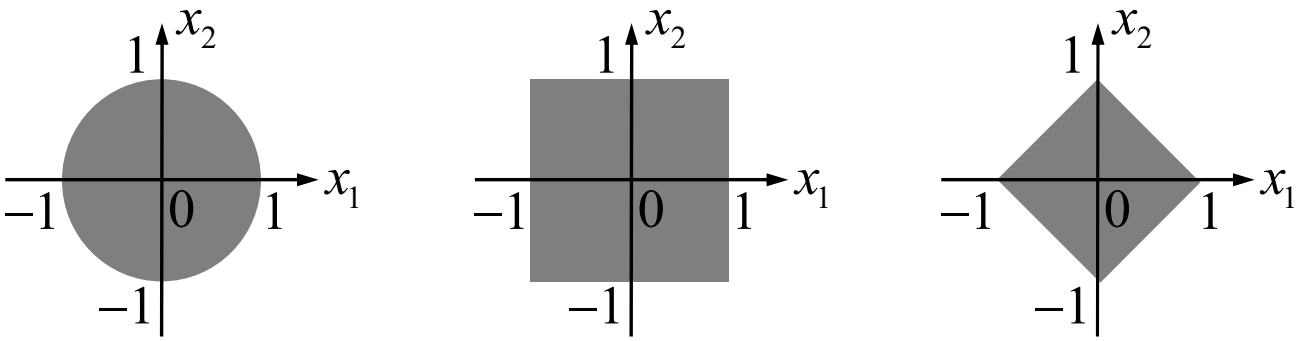


Figure: $B_1(0)$: 2-norm, ∞-norm, and 1-norm

Open balls do not contain their boundaries.

Open and Closed Sets

Definition: A set $\mathcal{S} \subseteq \mathbb{R}^n$ is open if for every $x \in \mathcal{S}$ there exists $\delta > 0$ such that $B_\delta(x) \subseteq \mathcal{S}$.

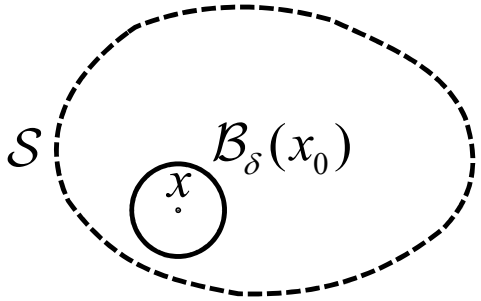


Figure: Open set $\mathcal{S}$

Definition: A set $\mathcal{S} \subseteq \mathbb{R}^n$ is closed if $\mathcal{S}^c = \mathbb{R}^n \setminus \mathcal{S}$ is open.

Fact: If $\mathcal{S}$ is closed then the limit of every convergent sequences formed from points in $\mathcal{S}$ are also in $\mathcal{S}$.

Examples

- $[0, 1]$ is closed, $(0, 1)$ is open
- $[0, 1)$ is neither open nor closed

Open and Closed Sets

Definition: Given a set $\mathcal{S} \subseteq \mathbb{R}^n$, its interior, denoted by int $\mathcal{S}$, is the largest open set contained in $\mathcal{S}$.

Definition: Given a set $\mathcal{S} \subseteq \mathbb{R}^n$, its closure, denoted by $\overline{\mathcal{S}}$, is the smallest closed set containing $\mathcal{S}$.

Definition: Given a set $\mathcal{S} \in \mathbb{R}^n$, its boundary, denoted by $\partial\mathcal{S}$, is

$$\partial\mathcal{S} = \overline{\mathcal{S}} \setminus \text{int}\, \mathcal{S}$$

Examples

- $\mathcal{S} = [0, 1)$, int $\mathcal{S} = (0, 1)$, $\overline{\mathcal{S}} = [0, 1]$, $\partial\mathcal{S} = \{0, 1\}$

Convergence

Definition: A sequence of points $\{x_k\}_{k\in\mathbb{N}} \subset \mathbb{R}$ converges to $x^* \in \mathbb{R}$ if $\forall \epsilon > 0$, $\exists k^* = k^*(\epsilon) \in \mathbb{N}$ such that $||x^* - x_{k^*}|| \leq \epsilon, \forall k \geq k^*$.

Notation: $\lim_{k\to\infty} x_k = x^*$ or $x_k \to x^*$ as $k \to \infty$.

Note that $\mathbb{N}$ is the set of natural numbers.

Note that all norms in $\mathbb{R}^n$ are equivalent, convergence in one norm implies convergence in another norm.

Compactness

Definition: A set $\mathcal{S} \subset \mathbb{R}^n$ is bounded if there exists $M < \infty$ such that $||x|| < M$, $\forall x \in \mathcal{S}$. Alternatively, a set $\mathcal{S} \subset \mathbb{R}^n$ is bounded if $\exists \delta > 0$ such that $||x - y|| < \delta$, $\forall x, y \in \mathcal{S}$.

Definition: A set $\mathcal{S} \subset \mathbb{R}^n$ is compact if it is both **closed** and **bounded**. This means that every sequence formed from elements of this set contains a subsequence which converges to an element in this set.

Continuity

Definition: A function $f : \mathcal{D} \to \mathbb{R}$, $\mathcal{D} \subseteq \mathbb{R}^n$ is continuous at $x_0 \in \text{int}\, \mathcal{D}$ if $\forall \epsilon > 0$, $\exists \delta = \delta(\epsilon) > 0$ such that

$$|f(x) - f(x_0)| < \epsilon, \quad \forall x \in \mathcal{D} \cap \mathcal{B}_\delta(x_0)$$

Fact: Continuous functions preserve convergence, $x_k \to x^*$ as $k \to \infty$ implies $f(x_k) \to f(x^*)$, i.e., $\lim_{k\to\infty} f(x_k) = f\left(\lim_{k\to\infty} x_k\right) = f(x^*)$.

Exercise

Let $f(x) = 2x^2 + 1$ and $x \in \mathbb{R}$. Show that f is continuous on $\mathbb{R}$.

Exercise

Let $f(x) = ||x||$ and $x \in \mathbb{R}^n$. Show that f is continuous.

Lipshitz Continuity

Definition: A function $f : \mathcal{D} \to \mathbb{R}, \mathcal{D} \subseteq \mathbb{R}^n$ is Lipshitz continuous if $\exists L > 0$ such that $|f(x) - f(y)| \leq L||x - y||, \forall x, y \in \mathcal{D}$.

Remark: A Lipshitz continuous function is continuous and differentiable almost everywhere.

Remark: A Lipshitz continuous function over an open set $\mathcal{D}$ is continuous over $\mathcal{D}$.

Minimum and Minimizer

Definition: Suppose $x^* \in \mathcal{D} \subseteq \mathbb{R}^n$ and $f(x^*) \leq f(x)\ \forall x \in \mathcal{D}$. Then, x^* is a (global) minimizer of f on $\mathcal{D}$ and $f(x^*)$ is the (global) minimum of f on $\mathcal{D}$.

Definition: Suppose $x^* \in \mathcal{D} \subseteq \mathbb{R}^n$ and $\exists \epsilon > 0$ such that $f(x^*) \leq f(x)$, $\forall x \in \mathcal{D} \cap \mathcal{B}_\epsilon(x^*)$. Then, x^* is a (local) minimizer of f on $\mathcal{D}$.

Fact: x^* is a global minimizer $\Rightarrow$ x^* is a local minimizer.

We often say x^* is a local minimizer of f or x^* is a local minimum point of f instead of f has a local minimum in x^* and so on. The minimizer is a point $x^* \in \mathcal{D}$, the minimum is the corresponding value $f(x^*)$.

Minimum and Minimizer

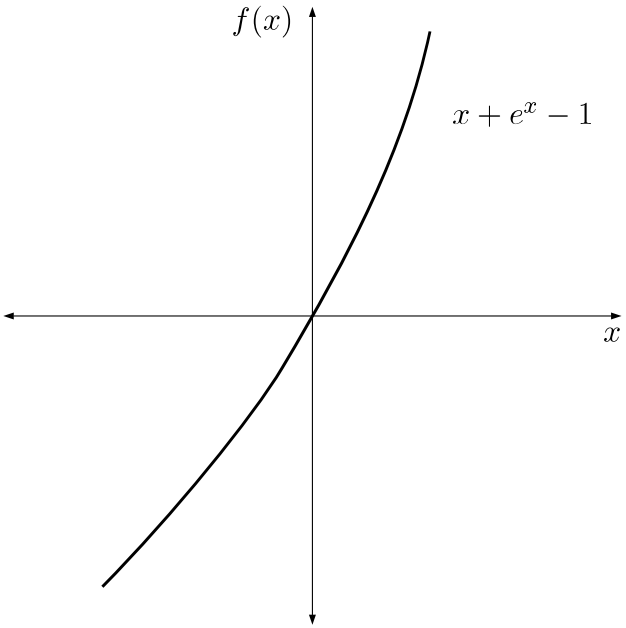
Definition: Suppose $x^* \in \mathcal{D} \subseteq \mathbb{R}^n$ and $f(x^*) < f(x), \forall x \in \mathcal{D}, x \neq x^*$. Then, x^* is a strict (global) minimizer of f on $\mathcal{D}$ and $f(x^*)$ is the minimum of f on $\mathcal{D}$.

Definition: Suppose $x^* \in X \subset \mathbb{R}^n$ and $\exists \epsilon > 0$ such that $f(x^*) < f(x)$, $\forall x \in \mathcal{D} \cap \mathcal{B}_\epsilon(x^*)$. Then, x^* is a strict (local) minimizer of f on $\mathcal{D}$.

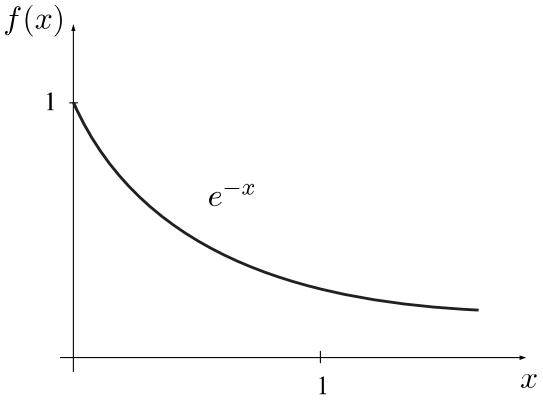
Fact: x^* is a strict global minimizer $\Rightarrow$ x^* is a strict local minimizer.

Note that a strict local minimizer is also referred to as an isolated minimizer.

Example

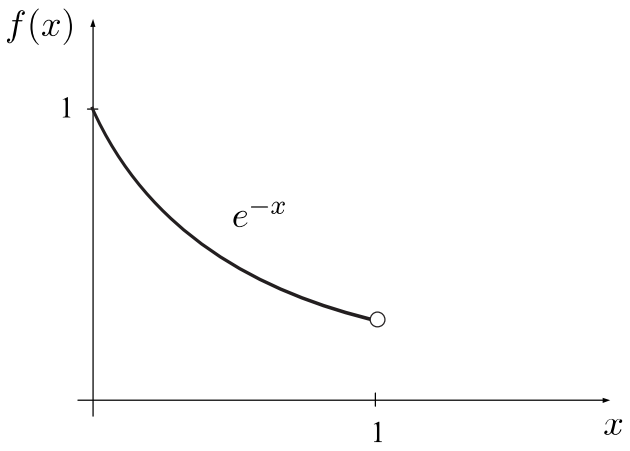


This strictly convex function, which has a closed, unbounded domain, has no lower bound and thus no global minimizer.

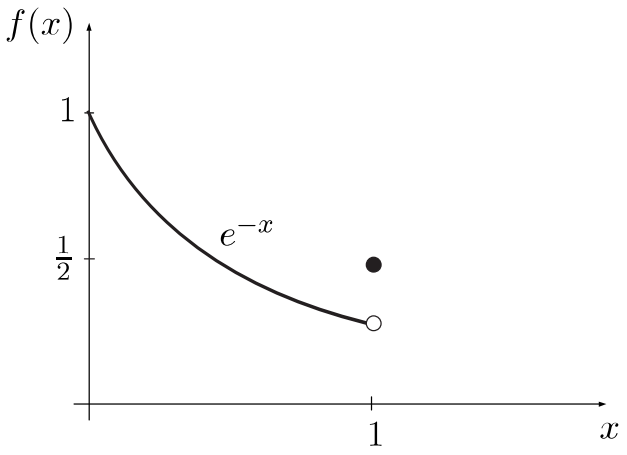


This strictly convex function, which has a closed, unbounded domain, has a lower bound but no global minimizer.

Example

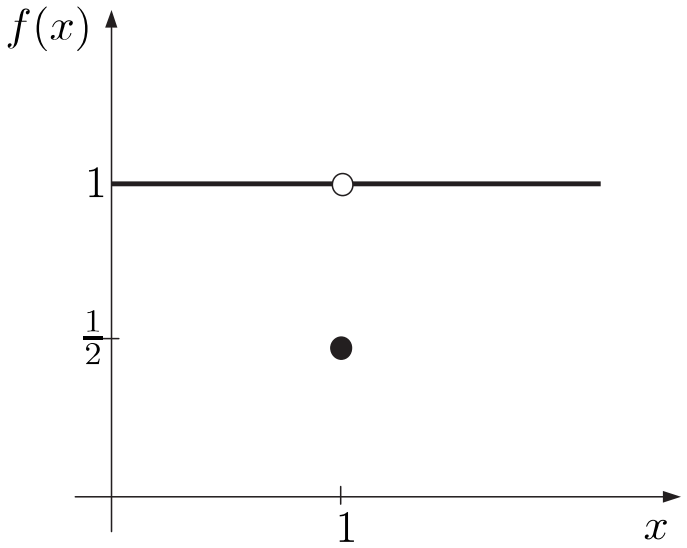


This strictly convex function, which has a bounded domain that is not closed, has no global minimizer.



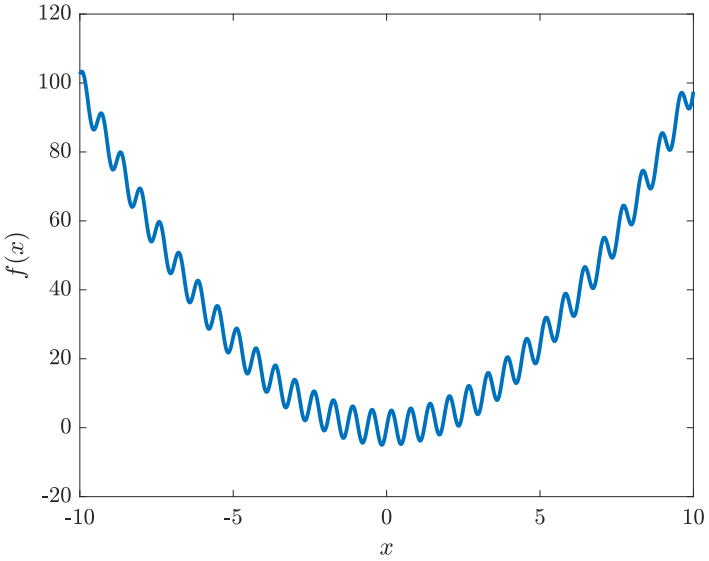
This strictly convex function, which is not continuous and has a compact domain, has no global minimizer

Example



This function, which is not continuous, has a strict global minimizer that is not an isolated local minimizer.

Example



This function, which is continuous, has multiple local minimizers.

Equivalence of Maximization and Minimization

Definition: The set of all minimizers of a given function f on a set $\mathcal{D} \subseteq \mathbb{R}^n$ is denoted by $\underset{x \in \mathcal{D}}{\operatorname{argmin}} f(x)$.

Definition: The set of all maximizers of a given function f on a set $\mathcal{D} \subseteq \mathbb{R}^n$ is denoted by $\underset{x \in \mathcal{D}}{\operatorname{argmax}} f(x)$.

Fact: $\underset{x \in \mathcal{D}}{\operatorname{argmin}} f(x) = \underset{x \in \mathcal{D}}{\operatorname{argmax}} (-f(x))$.

Fact: If $\mathcal{D}_1 \subseteq \mathcal{D}_2 \subseteq \mathbb{R}^n$, then $\arg\min_{x \in \mathcal{D}_1} f(x) \subseteq \arg\min_{x \in \mathcal{D}_2} f(x)$.

Weierstrass Theorem

Theorem: If $f : \mathcal{D} \to \mathbb{R}$ is a continuous function defined on a compact set $\mathcal{D} \subset \mathbb{R}^n$, then there exists $x^*, y^* \in \mathcal{D}$ such that

$$f(x^*) = \inf_{x \in \mathcal{D}} f(x), \quad f(y^*) = \sup_{x \in \mathcal{D}} f(x)$$

That is, $f(x^*) \leq f(x) \leq f(y^*)$

f is continuous, $\mathcal{D}$ is bounded $\Rightarrow$ the set of values of $f(x)$ on $\mathcal{D}$ is bounded. Hence the set of values of $f(x)$ on $\mathcal{D}$ has an infimum and a supremum, $f(x^*)$ and $f(y^*)$

Weierstrass theorem shows that $\exists x^*, y^* \in \mathcal{D}$ at which the infimum and the supremum are achieved. This means that a minimizer and a maximizer exist; it is a very strong theorem.

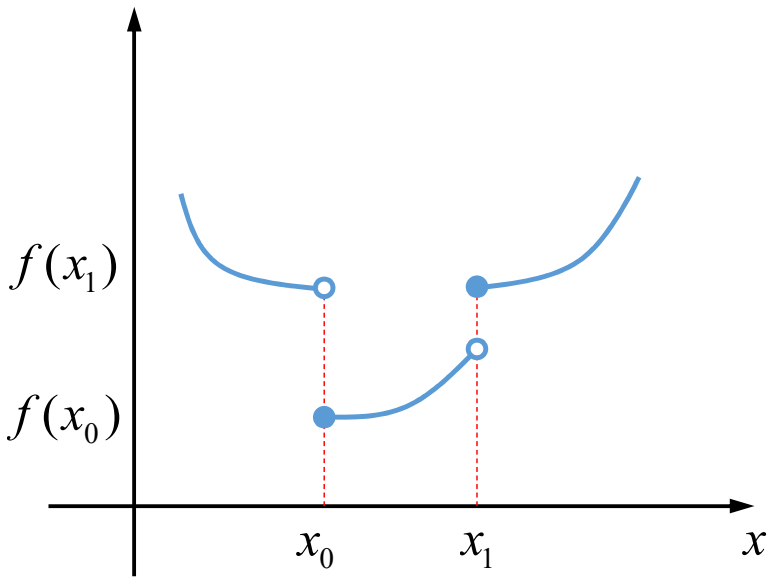
In Weierstrass theorem, the requirement of compactness of $\mathcal{D}$ can be replaced by the requirement of the compactness of

$$\mathcal{D}' = \mathcal{D} \cap \{x : f(x) \leq f(\tilde{x})\}, \quad \tilde{x} \in \mathcal{D}$$

Lower Semicontinuity

Definition: A function $f : \mathcal{D} \to \mathbb{R}, \mathcal{D} \subseteq \mathbb{R}^n$ is lower semicontinuous at $x_0 \in \text{int}\mathcal{D}$ if $\forall \epsilon > 0, \exists \delta = \delta(\epsilon)$ such that

$$f(x) \geq f(x_0) - \epsilon, \quad \forall x \in \mathcal{D} \cap \mathcal{B}_\delta(x_0).$$



The function is lower semicontinuous at x_0 but not lower semicontinuous at x_1

Generalized Weierstrass Theorem

Suppose that a function f is lower semicontinuous at all points $x \in \mathcal{D} \subset \mathbb{R}^n$, where $\mathcal{D}$ is compact (closed and bounded). Then, a minimizer exists on $\mathcal{D}$, i.e., there exists $x_0 \in \mathcal{D}$ such that $f(x_0) = \inf_{x \in \mathcal{D}} f(x)$.